

# Krishna soni

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[linkedin](#) | [github](#) | [leetcode](#)

## Summary

AI/ML engineer specializing in Retrieval-Augmented Generation and agentic systems, with hands-on experience building production-grade RAG pipelines, multi-agent research frameworks, and LangGraph-based recruiting workflows. Focused on applying evaluation-driven development to move beyond prototype-stage AI systems toward measurable, deployable products.

## Education

University of Petroleum and Energy Studies, Bachelor of Technology in ECE

Aug 2025 – May 2029

- CGPA: 7.85

## Skills

**Languages & Frameworks:** Python (FastAPI), C++ , JavaScript, TypeScript (Node.js)

**Databases & Caching:** PostgreSQL, MySQL, MongoDB, Redis, ChromaDB, Qdrant (Vector DBs)

**Infrastructure & DevOps:** Docker, AWS (EC2/S3), Git, CI/CD, MCP (Model Context Protocol), Streamlit Cloud

**Concepts:** Retrieval-Augmented Generation (RAG), Hybrid Search, Agentic Workflows, Neural Networks, Supervised/Unsupervised Learning, Fine-Tuning, Feature Engineering, Evaluation Metrics (recall@k, MRR, nDCG)

**AI/ML Frameworks:** PyTorch, TensorFlow, LangChain (AI Agents), LangGraph, Scikit-learn, Deep Learning, NLP

## Projects

### AI Research Framework

[Live Link](#) | [GitHub](#)

- Designed a cooperative multi-agent system (**agents.py**) with independent LLM configs and a stateful pipeline (**pipeline.py**) separating web discovery, synthesis, and report generation; exposed agent tools via a decorator-based system (**tool.py**) with error handling.
- Exposed core agent pipeline functionality via a FastAPI REST API, decoupling backend logic from the Streamlit frontend for independent testing and reuse.
- Shipped an interactive Streamlit dashboard with deterministic dependency management using the **uv** package engine, and implemented Streamlit-native authentication with **bcrypt** and **SQLAlchemy**.

### AI Recruiting Matching Agent

[GitHub](#)

- Designed the system architecture for an agentic LangGraph pipeline — multi-node workflow (JD parsing, requirement extraction, hybrid RAG resume search via ChromaDB dense + BM25 sparse retrieval with Mistral embeddings, multi-round candidate ranking) plus a human-in-the-loop feedback node.
- Built an evaluation harness measuring retrieval quality against hand-labeled ground truth, achieving **0.92 recall@3** and **0.90 nDCG@3** across 6 test queries spanning in-domain and out-of-domain resumes.

### AI Video Assistant with RAG Pipeline

[GitHub](#)

- Engineered a multimodal pipeline that ingests raw video, extracts audio, and generates transcripts, then answers queries via a RAG workflow using Mistral AI with transcript chunking to stay within context window limits.

## Open Source

### Mem0

[GitHub](#)

- Contributed a merged pull request to Mem0, an open-source memory layer for LLM applications, improving the project's codebase through the GitHub review process with project maintainers.

## Certifications

- **Supervised Machine Learning** : [\[View Certificate\]](#)
- **Software Engineer Intern Certification** — HackerRank [\[View Certificate\]](#)
- **Advanced Learning Algorithms** — DeepLearning.AI & Stanford University [\[View Certificate\]](#)
- **HackerRank Certificates** — SQL (Advanced) [\[View Certificate\]](#)